

Innovation Declaration

We used LLM tools as a coding partner: they helped draft functions, tidy notebooks, and sanity-check edge cases. But the main value of the work was not "ask an LLM for a trading strategy and run it". The important choices came from repeatedly looking at the data, refusing easy answers, and deciding what evidence would still look credible to a sceptical marker.

1. We did not force the stylised fact to look nicer.

A generic LLM answer would probably start from the textbook claim that overnight returns dominate close-to-close returns and then build the story around that. Our data did not quite behave that way: the return identity reconciled cleanly, but the equal-weight close-to-close stream was larger because intraday returns were positive. We treated that as a result to disclose, not a bug to hide. That choice shaped the whole report: audit the arithmetic first, then be honest about where the supplied universe differs from the stylised fact.

2. We changed the target, not just the model wrapper.

Instead of chasing a more impressive black-box model, we kept the point-in-time feature set stable and changed the learning problem to a volatility-scaled overnight target: $\text{overnight_next} / (\text{vol20} / \sqrt{252})$. That was a human modelling judgement. Overnight labels are noisy, and raw high-volatility names can dominate a simple learner for the wrong reason. Scaling the label made the question closer to "which names have unusually good overnight payoffs per unit of recent risk?" The final expanding-window learner then used more historical evidence over time without adding opaque complexity.

3. We made promotion depend on uncomfortable evidence, not just the best Sharpe.

The final strategy was promoted only after a design / validation / internal-holdout split, full cost accounting, borrow-cost treatment, capacity checks at 50M / 250M / 1B, and feature ablations. The internal holdout Sharpe was much lower than validation, and the return/reversal ablation showed a real dependence on one feature group. We kept those facts in the submission. That is the part we think goes beyond an off-the-shelf LLM workflow: the strategy is not presented as magic, but as a controlled improvement with clear failure modes.

In short, the LLM helped us move faster, but the added value was the judgement about what not to over-optimize: do not massage the data story, do not solve a noisy target with unnecessary complexity, and do not promote a strategy unless the weak evidence is visible too.

Final strategy referenced in the submission: phase2_g5_05_expanding, 250M net Sharpe 1.445 over 2010-2024, with 2025-2026 left untouched for marker evaluation.